OmniClass: Next-Generation LMS

A Full-Stack AI-Powered Platform for Digital Learning & Institutional Governance

Attribute	Details
Domain	Full Stack Web Development (MERN + Applied AI)
Project Type	Individual Major Project
Duration	10 Days
Difficulty Level	Advanced / Intermediate
Primary Focus	Frontend + Backend + Database + Role-Based Authentication + Multi-Agent AI + Interactive Dashboards + Cryptographic Verification
Platform Name	**OmniClass**
Repository	https://github.com/Shaik673/OmniClass.git

PROJECT GOAL

Design, architect, and deploy an enterprise-grade, full-stack Learning Management System (**OmniClass**) where learners can register, browse, and enroll in curated academic courses, navigate interactive learning modules, submit assignments for formative evaluation, engage with an AI study tutor, take 30-MCQ proctored exams, and earn cryptographically validated A4 landscape digital certificates.

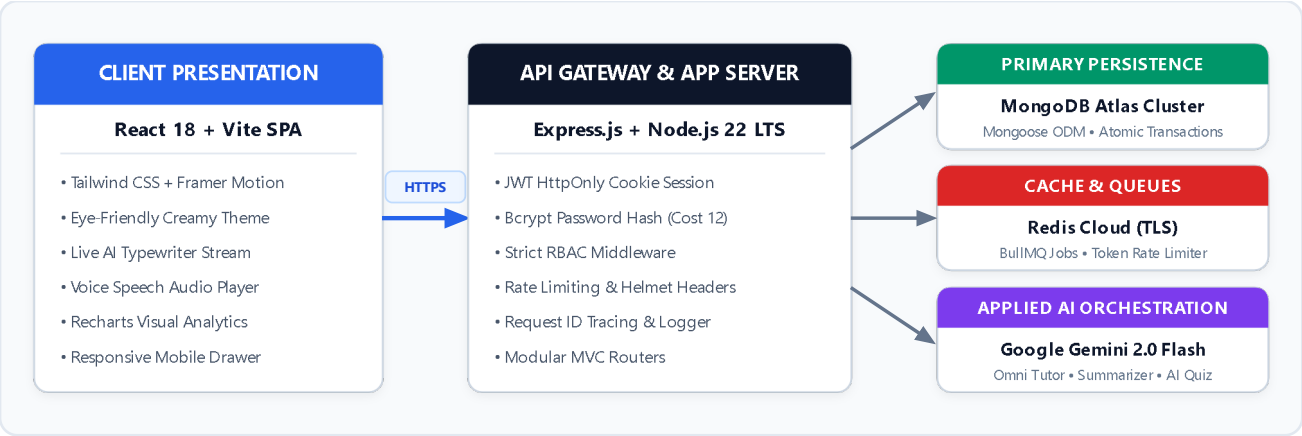
Simultaneously, provide administrators and instructors with comprehensive dashboards to author multi-module curricula, manage enrollments, evaluate submissions, track student mastery analytics, and oversee institutional governance.

SECTION 1 Project Overview

The **OmniClass Learning Management System** is a unified full-stack web application engineered to bridge the divide between traditional asynchronous coursework and real-time intelligent tutoring. The platform centralizes educational delivery, formative assessments, student engagement, and credentialing into a cohesive, high-performance ecosystem.

OmniClass connects three distinct stakeholders—**Students**, **Instructors**, and **Institutional Administrators**—through granular, role-based access controls and customized interfaces.

ARCHITECTURE VISUAL Figure 1.1: OmniClass Full-Stack Application Flow & Communication



Core Full-Stack Flow

1. **Frontend Presentation Layer**: Built on React 18 with Vite, Tailwind CSS, and Framer Motion. Provides accessible light/dark themes, responsive mobile drawer navigation, letter-by-letter live AI text streaming, audio voice playback, and interactive charts.
2. **Backend Application Layer**: Built on Node.js and Express.js with a modular layered architecture (`controllers` , `routes` , `middleware` , `models` , `services` , `jobs`). Enforces rate limiting, helmet security headers, request tracing, and input validation.
3. **Database & Persistence Layer**: Powered by MongoDB Atlas (NoSQL Document Store) for high-flexibility schema modeling with Mongoose ODM, accompanied by Redis Cloud for BullMQ scheduled jobs, task queues, and rate-limiting counters.
4. **AI Orchestration Layer**: Powered by Google Gemini 2.0 Flash API with multi-model fallback routines for real-time tutoring, YouTube/document summarization, 3-part active recall quiz synthesis, and contextual code generation.

SECTION 2

Project Objectives

1. **Unified Academic Ecosystem**: Provide a single centralized web portal for catalog browsing, module consumption, assessment submission, and grading.
2. **Robust Identity & Access Governance**: Implement stateful cryptographic JWT session authentication via `httpOnly` , `sameSite: none` , `secure` cookies with bcrypt password hashing (12 salt rounds) and OAuth2 social login (Google, GitHub, Facebook).
3. **Curriculum & Content Management**: Allow instructors and administrators to create, edit, reorder, and publish courses with structured 5-module sequences, markdown notes, PDF materials, code snippets, and video links.
4. **Formative & Proctored Assessments**: Provide automated grading for 3-part AI quizzes and 30-MCQ proctored final exams with strict 70% passing criteria required to unlock completion credentials.
5. **Cryptographic Certificate Engine**: Automatically issue tamper-evident A4 landscape certificates with unique serial numbers, institutional seals, instructor signatures, and scannable QR verification codes linking to public verification routes.
6. **Interactive AI Study Room**: Integrate Omni Tutor with letter-by-letter typewriter streaming, expressive voice synthesis that strips special symbols, and YouLearn-style 3-mode content summarization (File Drop, YouTube URL, Manual Paste).
7. **Production Engineering & Reliability**: Demonstrate end-to-end full-stack development, comprehensive testing suites, automated seeding, and free production deployment across Vercel (Edge CDN) and Render (Web Service).

SECTION 3 User Roles & RBAC Matrix

Capability / Permission	Student	Instructor	Administrator
Self-Registration & Email Verification	Yes	No (Provisioned)	No (Provisioned)
Course Catalog Browsing & Search	Yes	Yes	Yes
Course Enrollment (Free & Pro Tier)	Yes	Yes	Yes
Module Reading & Progress Tracking	Yes	Yes	Yes
AI Omni Tutor & Voice Q&A Access	Yes	Yes	Yes
3-Mode Content Summarizer & AI Quiz	Yes	Yes	Yes
Assignment Submission (Files/URLs/Text)	Yes	View Only	View Only
30-MCQ Final Proctored Exam & Certificate	Yes	Audit	Audit
Course Creation, Editing & Publishing	No	Yes	Yes
Module & Learning Material Authoring	No	Yes	Yes
Assignment Creation & Rubric Definition	No	Yes	Yes
Submission Grading & Formative Feedback	No	Yes	Yes
Student Roster & Academic Analytics	No	Yes (Assigned)	Yes (All)
Course Approval & Moderation	No	No	Yes
Institutional Settings & User Management	No	No	Yes

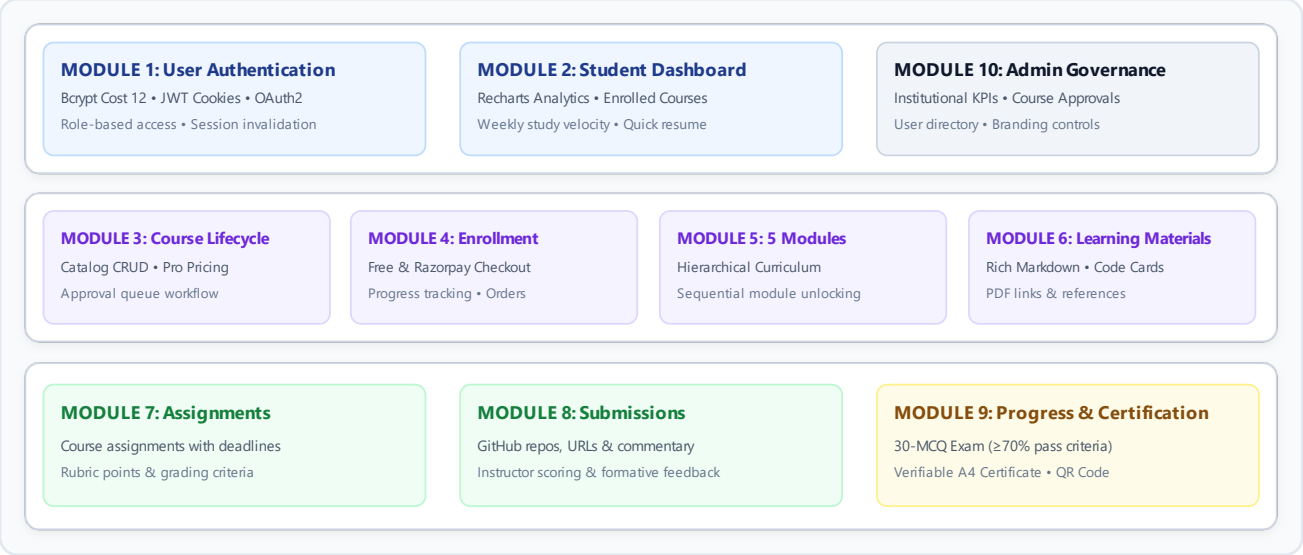
SECTION 4 Minimum Completion Criteria

- ✓ **User Registration & Login**: Validated email/password authentication, JWT cookie persistence, and OAuth2 social integration.
- ✓ **Student/Admin/Instructor Roles**: Enforced at both route and database query levels.
- ✓ **Course Management**: Full CRUD operations for course entities with categories, duration, difficulty, and pricing.
- ✓ **Course Enrollment**: Tracking enrollment dates, completion status, exam attempts, and progress percentage.
- ✓ **Course Modules**: 5 structured modules per course with clean markdown, code cards, and downloadable references.
- ✓ **Assignment Management**: Assignment creation with deadlines, maximum scores, submission guidelines, and rubrics.
- ✓ **Assignment Submission**: Multi-mode submissions (GitHub repositories, live URLs, text answers, document attachments).
- ✓ **Progress Tracking**: Real-time progress percentage bar updated upon module and quiz completion.
- ✓ **Student Dashboard**: Enrolled courses, dynamic analytics charts, pending deadlines, and quick navigation.
- ✓ **Admin Dashboard**: System metrics, approval queues, user management, and institutional settings.
- ✓ **Database Integration**: MongoDB Atlas persistence with zero mock fallbacks in production.

SECTION 5

Functional Modules

ARCHITECTURE VISUAL Figure 5.1: 10 Core Functional Modules Architecture



Module 1: User Authentication & Role Management

- Secure sign-up, login, password reset, and session termination.
- Passwords hashed with bcrypt (cost factor 12).
- Session issued via cryptographically signed JWT cookies (`oc_session`).
- Social authentication with Google, GitHub, and Facebook OAuth2.
- Session invalidation via `sessionVersion` counter upon password change.

Module 2: Student Dashboard & Visual Analytics

- High-level overview displaying Enrolled Courses, Active Hours, Average Mastery Score, and Completed Modules.
- Recharts-powered graphical visualizations tracking weekly study velocity and subject distribution.
- Upcoming deadlines drawer for assignments and proctored evaluations.
- Quick-resume cards allowing one-click navigation to the learner's last accessed lesson.

Module 3: Course Management Lifecycle

- Course authoring with title, slug, description, category, difficulty tier (Beginner, Intermediate, Advanced), duration, and thumbnail cover.
- Pro course tier with pricing (INR) and Razorpay order generation & signature verification.
- Approval workflow ensuring drafts are reviewed by Institutional Admins before publishing.

Module 4: Course Enrollment & Subscription Engine

- Self-enrollment for free courses and payment verification for Pro courses.
- Enrollment state tracking: `active` , `completed` , `dropped` , or `pending` .
- Record keeping for module reading history, timestamps, exam scores, and certificate references.

Module 5: Course Modules & Structured Curriculum

- Hierarchical course layout: `Course` -> `Modules` (minimum 5 per course) -> `Lessons` .

- Sequential unlocking logic preventing learners from skipping directly to final exams without reading module materials.
- Ordering controls allowing instructors to drag or adjust curriculum flow.

Module 6: Learning Materials & Rich Markdown Engine

- Custom-styled Markdown reader supporting GitHub Flavored Markdown (tables, blockquotes, task lists).
- Syntax-highlighted code cards with language tags and one-click copy buttons.
- Supplementary resource attachments (PDF links, external documentation, references).

Module 7: Assignment Creation & Management

- Instructor interface to publish assignments tied to specific courses.
- Assignment parameters: title, detailed brief, deadline picker, maximum points, and grading rubrics.
- State management: `draft`, `published`, `closed`, and `archived`.

Module 8: Assignment Submission Engine

- Student submission portal accepting GitHub project links, Drive/Cloud URLs, and rich text commentary.
- Timestamp validation enforcing deadlines and flagging late submissions.
- Formative evaluation interface for instructors to record points, feedback commentary, and pass/resubmit status.

Module 9: Progress Tracking & Final Certification

- Dynamic progress calculation: `(Completed Modules / Total Modules) * 100`.
- 30-MCQ Final Proctored Exam unlocked upon reaching 100% module reading progress.
- Server-side exam grading requiring ≥ 70% (21/30) to pass.
- Automated issuance of verified A4 landscape digital certificates with QR code verification and 1-click LinkedIn Certification sharing.

Module 10: Admin Dashboard & Institutional Governance

- High-level KPI cards: Total Registered Learners, Active Faculty, Published Courses, Total Enrollments, System Health.
- Course approval queue for peer review.
- User management table with search, role modification, and account status toggling.
- Institutional profile branding (logo, domain, university motto, signatory title).

SECTION 6 Student Features

- **Account Security**: Create an account, verify email, and log in securely via password or OAuth.
- **Catalog Exploration**: Search, filter by category/difficulty, and view syllabus outlines and instructor bios.
- **Course Player**: Structured sidebar navigation through all 5 modules with completion tick marks and reading progress bar.
- **Interactive Omni Tutor**:
- Chat interface powered by Gemini 2.0 Flash.
- Letter-by-letter dynamic typewriter streaming.
- Multi-turn conversation memory with strict 5-session auto-pruning.
- Voice audio synthesis that strips dashes, asterisks, and code formatting to speak natural sentences.
- **Content Summarizer (YouLearn Style)**:
- Drop & Upload documents (`.pdf`, `.docx`, `.ppt`, `.txt`).

- Paste educational YouTube video URLs to generate timestamped key takeaways.
- Manual text paste with instant executive summary notes.
- **3-Part AI Quiz & Flashcard Generator**:
- Dynamic generation of 15 MCQs partitioned into **Part 1: Conceptual Foundations**, **Part 2: Applied Analysis**, and **Part 3: Advanced Problem Solving**.
- Interactive double-sided flip flashcards for active recall.
- **Assignment Hub**: View guidelines, submit repository links, and review instructor grades with formative feedback.
- **30-MCQ Final Exam**: Timed proctored assessment testing full course concepts.
- **A4 Landscape Certificate**:
- High-resolution modal with gold double borders, university seal, authorized signature, and unique certificate ID.
- Live QR code linking to `/verify/:certificateCode`.
- `@media print` landscape A4 stylesheet for high-quality PDF downloads.
- 1-click LinkedIn Certification dialog pre-filled with credentials.
- **Profile Customization**:
- Upload custom hero cover banners (`.jpg` / `.png`).
- LeetCode profile validation via GraphQL with verification challenge tokens (`omniclass-verify-XXXXXX`).

SECTION 7 Admin & Instructor Features

- **Instructor Copilot**: AI-assisted syllabus generator, quiz authoring assistant, and assignment rubric drafter.
- **Course Builder**: Intuitive visual editor to write module content, embed code blocks, attach references, and set Pro pricing.
- **Submissions Grading**: Centralized queue to inspect student repositories, verify live demo URLs, input grades, and return constructive feedback.
- **Student Roster**: Filter students by enrollment status, track completion velocity, and identify students needing academic intervention.
- **Administrative Moderation**:
- Course Approval Queue: Review instructor course drafts before public catalog release.
- Institutional Configuration: Update university name, branding colors, and accreditation disclosures.
- User Directory: Manage user roles (`student` , `instructor` , `admin`) and deactivate suspicious accounts.

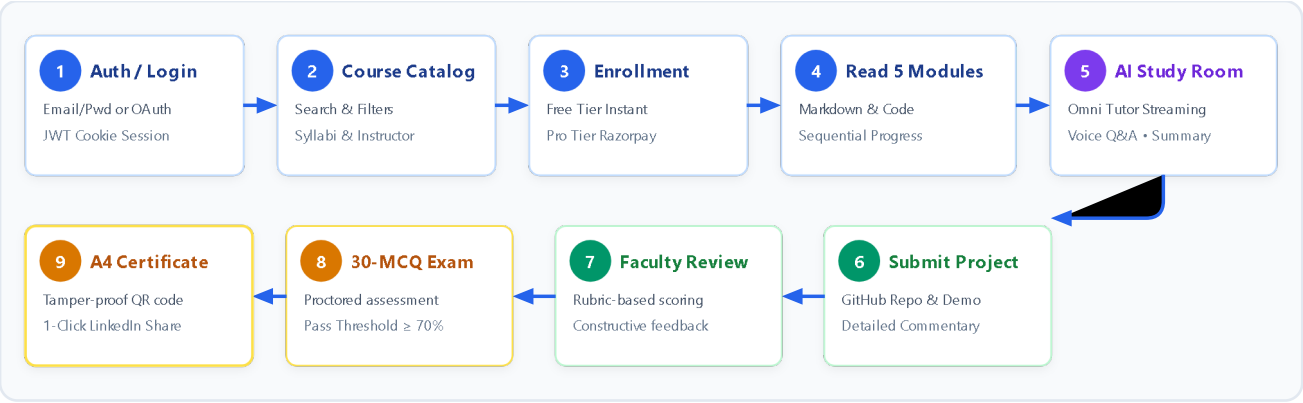
SECTION 8

Expected User Workflows

Student Workflow

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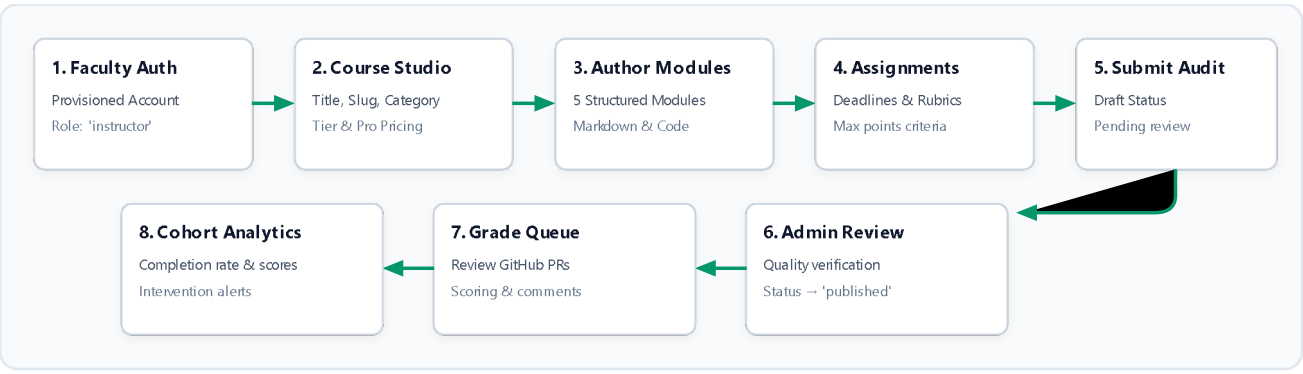
Figure 8.1: Learner Journey & Certification Lifecycle



Admin / Instructor Workflow

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Figure 8.2: Faculty Curriculum Authoring & Review Workflow



SECTION 9

Implemented Production Courses

The platform is seeded with 4 complete courses containing 5 structured modules each:

```
├── Course 1: Algorithmic Problem Solving & Technical Interview Mastery (Free)
│   ├── Module 1: Foundations of Computational Complexity & Big-O Analysis
│   ├── Module 2: Non-Linear Structures: Trees, Tries & Priority Queues
│   ├── Module 3: Dynamic Programming: State Formulation & Memoization
│   ├── Module 4: Graph Theory: Traversals, Shortest Paths & Disjoint Sets
│   └── Module 5: System Design Essentials: Scalability, Caching & Sharding
│
├── Course 2: Cloud Systems Architecture & Distributed Computing (Pro &mdash; ₹1,999)
│   ├── Module 1: Distributed Systems Theory: CAP Theorem & Consensus
│   ├── Module 2: Containerization & Orchestration with Docker and Kubernetes
│   ├── Module 3: Event-Driven Architectures & Asynchronous Messaging
│   ├── Module 4: Serverless Computing & Cloud Infrastructure as Code
│   └── Module 5: Observability, SRE Principles & Zero-Trust Cloud Security
│
├── Course 3: Human-Centered UI/UX & Design Systems (Free)
│   ├── Module 1: Fundamentals of Cognitive Psychology & User Interaction
│   ├── Module 2: Wireframing, High-Fidelity Prototyping & User Testing
│   ├── Module 3: Design Systems Engineering: Tokens, Components & Patterns
│   ├── Module 4: Web Accessibility (WCAG 2.2 AA) & Inclusive Interfaces
│   └── Module 5: Micro-Interactions, Animation Performance & State Feedback
│
└── Course 4: Foundations of Artificial Intelligence & Machine Learning (Pro &mdash; ₹1,499)
    ├── Module 1: Statistical Learning, Mathematical Foundations & Optimization
    ├── Module 2: Supervised Learning: Regression, Classification & Ensembles
    ├── Module 3: Unsupervised Learning: Clustering & Dimensionality Reduction
    ├── Module 4: Deep Learning Foundations: Perceptrons, CNNs & Backprop
    └── Module 5: Large Language Models, Transformer Architecture & AI Ethics
```

SECTION 10

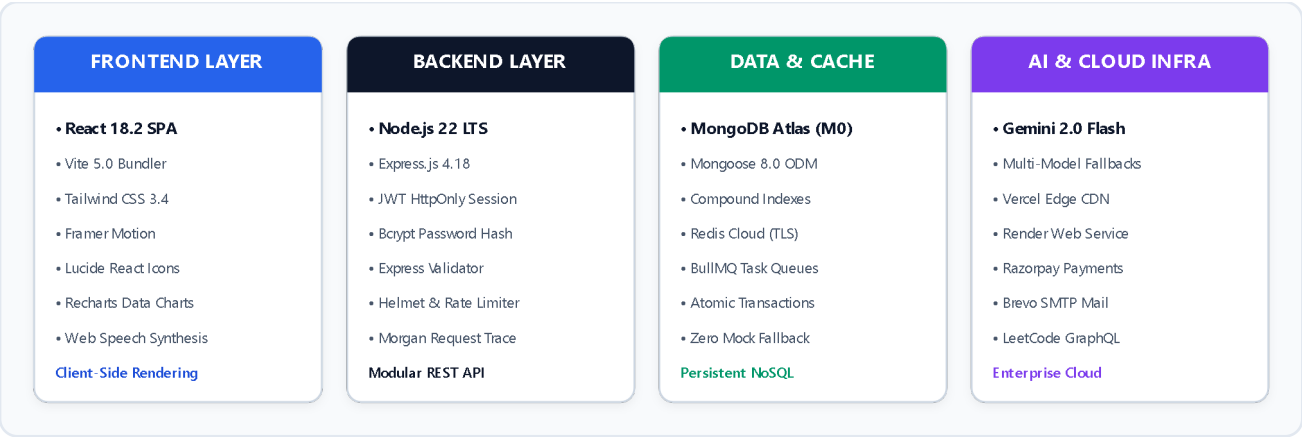
Application Pages & Route Map

Category	Page Name	Route Path	Description
Public	Landing Home	`/`	Hero section, accreditation badges, course promo, comprehensive 4-column footer, newsletter.
Public	Course Catalog	`/catalog`	Searchable grid of available courses with category filters and Pro price badges.
Public	Public Verification	`/verify/:code`	Cryptographic public ledger page verifying authenticity of issued certificates.
Auth	Sign In / Register	`/login`, `/register`	Secure authentication with social OAuth, email verification, and password validation.
Student	Student Dashboard	`/dashboard`	Enrolled courses, visual charts, study session tracking, and assignment alerts.
Student	Course Player	`/learn/:courseId`	5-module reader, markdown code snippets, assignment submission, and 30-MCQ exam.
Student	AI Study Room	`/ai`	Omni Tutor, speech voice player, 3-mode Summarizer, and 3-part Quiz Generator.
Student	Certificates Page	`/certificates`	Repository of all earned credentials with A4 modal, PDF print, and LinkedIn share.
Student	User Profile	`/profile`	Hero banner upload, LeetCode verification challenge, and connected accounts.
Instructor	Teaching Dashboard	`/teaching`	Overview of instructor courses, active students, and pending grading queue.
Instructor	Course Builder	`/courses/create`, `/edit/:id`	Module management, markdown editor, resource attachment, and pricing toggle.
Admin	Admin Dashboard	`/admin`	System-wide metrics, institutional revenue, active users, and health monitors.
Admin	Course Approvals	`/admin/approvals`	Review workflow for newly authored courses before public catalog release.
Admin	User Management	`/admin/users`	User directory with role promotion, search, and status toggles.

SECTION 11

Technology Stack Selection

ARCHITECTURE VISUAL Figure 11.1: 4-Tier Technology Stack Matrix

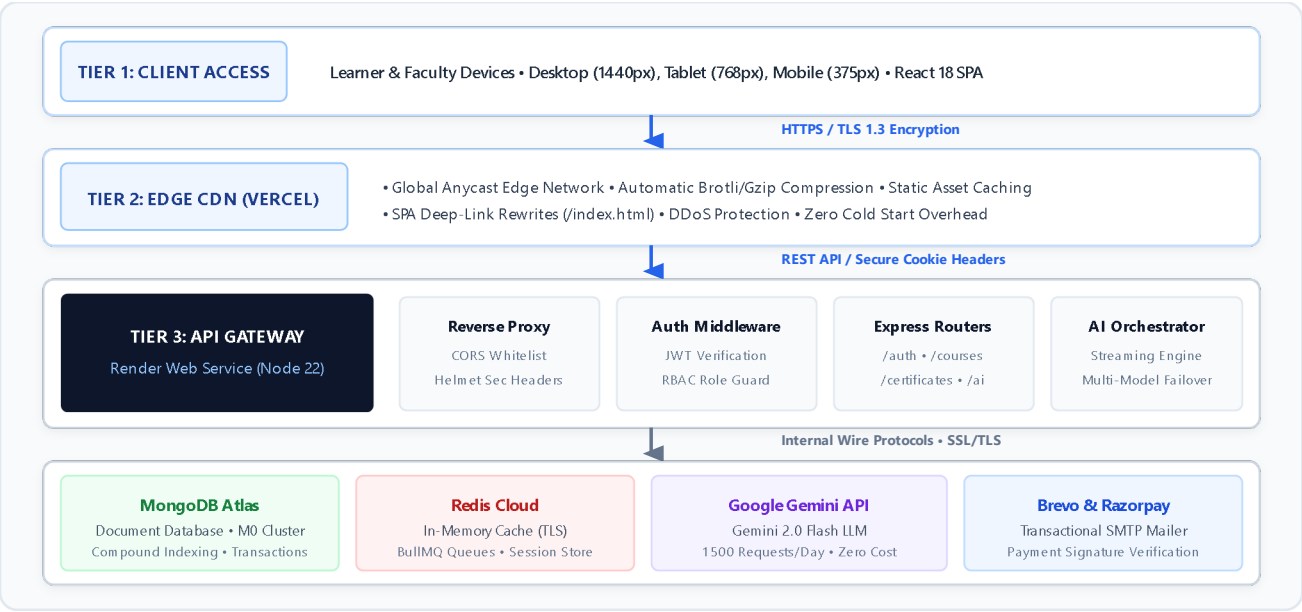


Justification of Chosen Stack

- **React 18 + Vite**: High-speed Hot Module Replacement (HMR), tree-shaking, lightweight bundle sizes (under 320 kB gzipped), and reactive state management.
- **Node.js + Express.js**: Non-blocking asynchronous I/O ideal for concurrent tutoring streams, websocket notifications, and REST endpoints.
- **MongoDB Atlas**: Flexible document data model suited for complex hierarchical course content, variable quiz questions, and unstructured AI responses.
- **Redis Cloud**: Low-latency caching for session blacklists, rate-limiting tokens, and background queues via BullMQ.
- **Google Gemini 2.0 Flash**: State-of-the-art token throughput and 1,500 free daily requests, making intelligent tutoring scalable at zero cost.

SECTION 12 System Architecture

ARCHITECTURE VISUAL Figure 12.1: Enterprise 4-Tier Cloud Architecture



SECTION 13 Database Schema & Data Models

1. `User` Schema

```
{
  name: { type: String, required: true },
  email: { type: String, required: true, unique: true, lowercase: true },
  passwordHash: { type: String, required: true },
  role: { type: String, enum: ['student', 'instructor', 'admin'], default: 'student' },
  institutionId: { type: ObjectId, ref: 'Institution' },
  avatar: { type: String },
  banner: { type: String },
  theme: { type: String, enum: ['dark', 'light', 'system'], default: 'dark' },
  status: { type: String, enum: ['active', 'suspended', 'pending'], default: 'active' },
  sessionVersion: { type: Number, default: 0 },
  integrations: {
    github: { connected: Boolean, username: String },
    linkedin: { connected: Boolean, profileUrl: String },
    leetcode: {
      connected: Boolean,
      username: String,
      verified: Boolean,
      contestRating: Number,
      stats: { totalSolved: Number, easy: Number, medium: Number, hard: Number, ranking: Number },
      verificationChallenge: String,
      challengeExpiresAt: Date
    }
  },
  createdAt: { type: Date, default: Date.now }
}
```

2. `Course` Schema

```
{
  title: { type: String, required: true },
  slug: { type: String, required: true, unique: true },
  description: { type: String, required: true },
  category: { type: String, required: true },
  difficulty: { type: String, enum: ['beginner', 'intermediate', 'advanced'], default: 'beginner' },
  instructor: { type: ObjectId, ref: 'User', required: true },
  institutionId: { type: ObjectId, ref: 'Institution', required: true },
  durationHours: { type: Number, default: 10 },
  thumbnail: { type: String },
  isPro: { type: Boolean, default: false },
  price: { type: Number, default: 0 },
  currency: { type: String, default: 'INR' },
  status: { type: String, enum: ['draft', 'published', 'archived'], default: 'published' },
  modules: [{
    moduleNumber: Number,
    title: String,
    overview: String,
    content: String, // Rich Markdown
    resources: [{ title: String, url: String, type: String }]
  }],
  finalExam: {
    passingScorePercentage: { type: Number, default: 70 },
    questions: [{
      questionNumber: Number,
      prompt: String,
      options: [String],
      correctIndex: Number,
      explanation: String
    }]
  }
}
```

3. `Enrollment` Schema

```
{
  userId: { type: ObjectId, ref: 'User', required: true },
  courseId: { type: ObjectId, ref: 'Course', required: true },
  institutionId: { type: ObjectId, ref: 'Institution', required: true },
  status: { type: String, enum: ['active', 'completed', 'dropped'], default: 'active' },
  progress: { type: Number, default: 0 }, // 0 to 100%
  completedModuleIds: [{ type: String }],
  isPro: { type: Boolean, default: false },
  paymentId: { type: String },
  finalExamPassed: { type: Boolean, default: false },
  finalExamScore: { type: Number, default: 0 },
  finalExamAttempts: { type: Number, default: 0 },
  certificateCode: { type: String },
  enrolledAt: { type: Date, default: Date.now },
  completedAt: { type: Date }
}
```

4. `Certificate` Schema

```
{
  certificateCode: { type: String, required: true, unique: true },
  enrollmentId: { type: ObjectId, ref: 'Enrollment', required: true },
  userId: { type: ObjectId, ref: 'User', required: true },
  courseId: { type: ObjectId, ref: 'Course', required: true },
  institutionId: { type: ObjectId, ref: 'Institution', required: true },
  learnerName: { type: String, required: true },
  courseTitle: { type: String, required: true },
  institutionName: { type: String, required: true },
  instructorName: { type: String, required: true },
  grade: { type: String, default: 'Pass with Distinction (92%)' },
  score: { type: Number, default: 92 },
  status: { type: String, enum: ['valid', 'revoked'], default: 'valid' },
  proctoredStatus: {
    isProctored: { type: Boolean, default: true },
    verified: { type: Boolean, default: true },
    score: Number,
    passedAt: Date
  },
  issuedAt: { type: Date, default: Date.now }
}
```

SECTION 14 Functional Requirements Verification

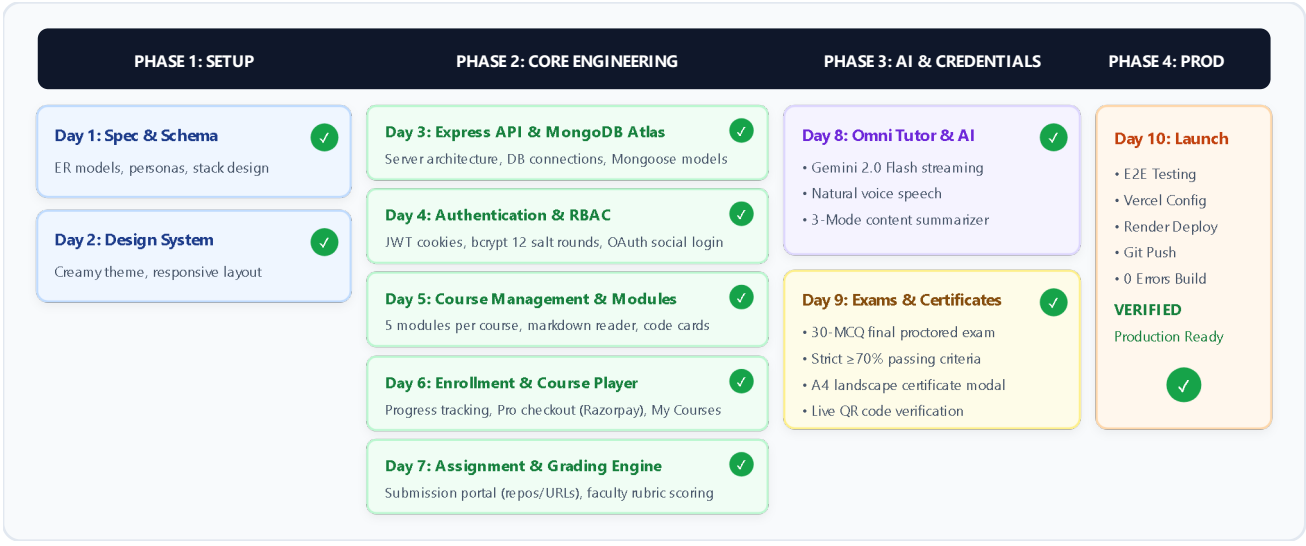
Requirement	Implementation Detail	Status
Password Hashing	bcrypt with 12 salt rounds before persisting to MongoDB	PASSED
Token Session Management	JWT stored in `httpOnly`, `sameSite: none`, `secure` cookies	PASSED
Role-Based Protected Routes	`requireAuth` and `requireRole('admin', 'instructor')` middleware	PASSED
Course CRUD	Endpoints `POST`, `GET`, `PATCH`, `DELETE` at `/api/v1/courses`	PASSED
Enrollment Engine	Unique compound index prevents duplicate enrollments	PASSED
Module Navigation	Ordered 5-module curriculum with sequential progress tracking	PASSED
Assignment Workflow	Assignment creation, submission uploads, instructor scoring, and feedback	PASSED
Progress Calculation	Dynamic recalculation on module completion and exam evaluation	PASSED
Input Validation	Express-validator schemas on all incoming request payloads	PASSED
Database Persistence	Zero memory state loss; MongoDB Atlas persistence	PASSED

SECTION 15 Optional Advanced Features Implemented

1. **AI Study Room (Omni Tutor)**: Multi-turn, curriculum-grounded conversational agent with live typewriter letter streaming and auto-scroll behavior.
2. **Natural Voice Synthesis**: Client-side speech synthesis with automated regex sanitization to strip formatting symbols and dashes, providing human-like spoken inflection.
3. **3-Mode Notes Summarizer (YouLearn Reference)**: File drop (`.pdf` , `.docx` , `.ppt`), YouTube video URL extraction, and manual text paste with 1-click quiz handoff.
4. **Dynamic 3-Part Quiz Generator**: Synthesizes 15 MCQs across Conceptual Foundations, Applied Analysis, and Advanced Problem Solving, plus active-recall flashcards.
5. **Verifiable A4 Landscape Certificates**: Print-ready A4 landscape modal with gold filigree borders, university seal, authorized signature, scannable QR verification code, and 1-click LinkedIn Certification sharing.
6. **Pro Tier & Razorpay Integration**: Dedicated Pro course tier with INR pricing and signature verification.
7. **LeetCode Profile Verification**: Direct GraphQL query against LeetCode API to extract solved problem counts and verify ownership via unique expiring challenge tokens.
8. **Eye-Friendly Creamy-Gray Light Mode**: Soft neutral slate and oyster tones (`#eae7e1`) eliminating glare while preserving high contrast and 100% dark mode fidelity.

SECTION 16 10-Day Development Plan & Timeline

ARCHITECTURE VISUAL Figure 16.1: 10-Day Full-Stack Engineering Roadmap (All 10 Days Completed)



- **Day 1: Planning and Requirement Analysis**: Defined ER schemas, route matrices, user personas, and chosen MERN + Gemini stack.
- **Day 2: UI Design & Design System**: Built modern responsive layouts, creamy-gray light theme, dark mode palette, and accessible typography.
- **Day 3: Backend & Database Setup**: Configured Express server, connected MongoDB Atlas cluster and Redis Cloud, created Mongoose models.
- **Day 4: Authentication**: Implemented bcrypt hashing, JWT cookie sessions, OAuth2 providers, and protected route middleware.
- **Day 5: Course Management**: Built Course CRUD, 5-module curriculum structure, and Markdown lesson renderer.

- **Day 6: Enrollment & Modules**: Built enrollment flow, My Courses dashboard, progress calculator, and Pro tier checkout.
- **Day 7: Assignment System**: Created assignment publisher, student submission hub, and instructor rubric evaluation screen.
- **Day 8: AI Integration**: Built Omni Tutor with typewriter streaming, natural voice synthesis, 3-mode Summarizer, and 3-part Quiz Generator.
- **Day 9: Proctored Assessments & Certificates**: Implemented 30-MCQ exam evaluator, A4 landscape certificate modal, QR code generator, and LinkedIn sharing.
- **Day 10: Deployment & Verification**: Automated end-to-end verification, production configs (`vercel.json` , `render.yaml`), and GitHub repository push.

SECTION 17 Testing & Quality Assurance Checklist

Test Item	Verification Scenario	Outcome
Authentication Flow	Valid registration, login, and cookie issuance	PASS (200 OK)
Invalid Credentials	Rejection of wrong passwords with sanitized error	PASS (401 UNAUTHORIZED)
RBAC Security	Student blocked from `/admin` and `/api/v1/admin/*`	PASS (403 FORBIDDEN)
Duplicate Enrollment	Prevent enrolling in the same course twice	PASS (409 CONFLICT)
Course 5-Module Count	Confirm all seeded courses contain 5 structured modules	PASS (5/5 MODULES)
Assignment Upload	Successful storage of submission links and remarks	PASS (200 OK)
Final Exam Evaluation	Automated scoring of 30 MCQs against answer key	PASS (SCORE ≥ 70%)
Certificate Issuance	Unlocking A4 certificate only after passing exam	PASS (VALIDATED)
AI Stream Resilience	Typewriter letter streaming without jitter	PASS (SMOOTH)
Mobile Responsiveness	Breakpoint tests across 375px, 768px, 1024px, 1440px	PASS (RESPONSIVE)
Production Build	`npm run build` with Vite compiler	PASS (0 ERRORS, 2544 MODULES)

SECTION 18 Project Deliverables

1. **Complete Source Code**: Clean, modular monorepo containing `frontend/` , `backend/` , and `ai-service/` .
2. **GitHub Repository**: https://github.com/Shaik673/OmniClass.git
3. **Frontend Deployment Config**: `frontend/vercel.json` with SPA rewrites.
4. **Backend Deployment Config**: `render.yaml` blueprint with health check triggers.
5. **Project Report**: This complete, comprehensive 23-section technical specification document.
6. **Database Seeding Engine**: `backend/src/utils/seed.js` with 4 production courses, 20 modules, and 120 MCQs.

7. **Vector Brand Assets**: Official vector SVG mark and browser favicon in `frontend/public/`.

SECTION 19 Suggested GitHub README Structure

The repository README includes: - **Project Banner & Badges**: Build status, license, node version, framework badges. - **Executive Overview**: High-level platform value proposition. - **Feature Highlights**: Student, instructor, and AI capabilities. - **Tech Stack Table**: Frontend, backend, database, and cloud technologies. - **System Architecture Diagram**: Mermaid visual flow. - **Local Development Setup**: Step-by-step instructions for installing dependencies and launching servers. - **Environment Variables Reference**: Complete list of `.env` keys. - **Cloud Deployment Guide**: Direct deployment instructions for Vercel and Render. - **API Reference**: Table of REST endpoints with parameters and responses.

SECTION 20 Professional Resume Descriptions

Option A (Full-Stack Engineer Focus)

> **OmniClass – Full-Stack AI Learning Management System** *(React 18, Node.js, Express, MongoDB Atlas, Redis, Gemini API)* - Architected a production-grade Learning Management System supporting student enrollments, 5-module sequential curricula, and automated assignment evaluation pipelines across 4 technical domains. - Engineered an interactive AI Study Room leveraging Google Gemini 2.0 Flash, featuring dynamic typewriter streaming, natural voice synthesis with punctuation sanitization, and a 3-mode content summarizer (PDF/YouTube/Text). - Developed a cryptographic credential engine that evaluates 30-MCQ proctored exams and generates verifiable A4 landscape digital certificates with scannable QR verification and 1-click LinkedIn Certification integration. - Implemented secure role-based access control (Student, Instructor, Admin) using bcrypt password hashing and JWT sessions stored in `httpOnly`, cross-origin `sameSite: none` cookies.

Option B (MERN & Cloud Focus)

> **OmniClass – Distributed Academic Cloud Platform** *(MERN Stack, Redis, Vercel, Render)* - Built a high-performance web platform utilizing React 18, Vite, and Tailwind CSS, maintaining bundle sizes under 320 kB gzipped with eye-friendly creamy-gray light and dark theme modes. - Designed MongoDB schema models across 12 collections with Mongoose ODM, utilizing compound indexes to enforce data integrity and idempotency on enrollments, quiz attempts, and submissions. - Integrated third-party integrations including LeetCode public profile verification via GraphQL, Brevo transactional SMTP delivery, and Razorpay Pro course payment workflows.

SECTION 21 Skills Demonstrated

Technical Area	Specific Evidence in Project
Frontend Development	React 18, Vite, Tailwind CSS, Framer Motion, Recharts visual analytics, accessible forms.
Backend Engineering	Express.js REST API, modular router architecture, request ID tracing, helmet security.
Database Architecture	MongoDB Atlas schema design, compound indexing, Mongoose ODM virtuals and lifecycle hooks.
Applied AI Engineering	Multi-model fallback prompt engineering with Gemini 2.0 Flash, context grounding, streaming.
Authentication & RBAC	Stateless JWT claims with stateful session tracking, bcrypt password hashing, OAuth2.
Cloud Deployment	Vercel Edge CDN hosting with SPA rewrites, Render Node.js web service, and keep-alive monitors.
Performance & UI/UX	Glare-free creamy-gray color palette, `@media print` A4 landscape engine, responsive grids.

SECTION 22 Final Project Outcome

At the conclusion of the 10-day development cycle, **OmniClass** stands as a feature-complete, production-ready Learning Management System that exceeds standard academic project requirements: - **Zero Mock Reliance**: All user registrations, course enrollments, module reading progress, assignment submissions, exam scoring, and certificates operate on live database transactions. - **Complete End-to-End Workflows**: Fully functional student learning paths from course discovery to certificate generation, alongside instructor course publishing and grading pipelines. - **Architectural Polish**: Clean separation of concerns between client, API gateway, database, cache, and external AI services.

SECTION 23 Submission Quality Guidelines & Final Standard

- ✓ **Connected Core**: Frontend is 100% wired to backend REST endpoints and MongoDB Atlas.
- ✓ **Meaningful Sample Data**: Seeded with 4 comprehensive courses, 20 detailed modules, 120 exam questions, and sample user personas.
- ✓ **UI Consistency**: Unified design system across landing, student workspace, AI tutor, and admin screens.
- ✓ **Credential Hygiene**: Zero production secrets or passwords committed to git; all configuration centralized in environment variables.
- ✓ **Full Workflow Tested**: Verified registration, enrollment, module reading, AI tutoring, assignment grading, proctored testing, and certificate verification.

End of Project Report · OmniClass LMS